

Maternal RSV Vaccination, Infant Nirsevimab, or Both: Interim Analysis of a Randomized Trial

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BACKGROUND: Although both maternal respiratory syncytial virus (RSV) prefusion F vaccination (RSVpreF) and infant nirsevimab immunization have been approved for the prevention of RSV lower respiratory tract infections, the 2 have not been evaluated in a single study, and their sequential administration has not been studied systematically.

METHODS: We performed a prospective, randomized, open-label, phase 4 study at 8 US sites of mother-infant pairs randomized 1:1:1:1 during pregnancy: maternal RSVpreF vaccine alone, maternal RSVpreF vaccine/infant nirsevimab at birth, maternal RSVpreF vaccine/infant nirsevimab at 3 months, or infant nirsevimab alone at birth. We are following the mother-infant pairs for 12 months to ascertain safety, infant tolerability, and the magnitude and durability of RSV-A and -B neutralizing antibodies (nAbs). We report interim data from September 19, 2024, to May 15, 2025, including 4-month infant follow-up.

RESULTS: In total, 181 mothers were enrolled. Both products alone and in combination were safe. No related serious adverse events were observed in mothers or infants. Nirsevimab was well tolerated, and all local and systemic reactogenicity was mild to moderate in severity. RSVpreF vaccination boosted maternal RSV-A nAb titers 17.35-fold at the time of delivery, and titers were durable through 3 months postdelivery. The geometric mean transfer ratio of RSV-A nAbs was higher than 1.3 and similar across groups. RSV nAbs were highly elevated in infants at 6 weeks and 3 months, irrespective of group, with modest differences in waning.

CONCLUSIONS: Maternal RSVpreF vaccine and infant nirsevimab administration, either alone or in combination, were safe and provided high RSV nAb titers in infants through interim follow-up.



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WHAT'S KNOWN ON THIS SUBJECT: Although both maternal respiratory syncytial virus (RSV) vaccination and infant nirsevimab immunization have been approved for the prevention of infant RSV lower respiratory tract infections, the 2 products have not been evaluated in a single study, and their sequential administration has not been studied systematically.

WHAT THIS STUDY ADDS: Maternal RSV prefusion F vaccination (RSVpreF) and infant nirsevimab administration, either alone or in combination, were safe and provided high RSV-A and -B neutralizing antibody titers in infants that persisted through 3 months of age.

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INTRODUCTION

Respiratory syncytial virus (RSV) is a leading cause of lower respiratory tract disease (LRTD) and hospitalization in infants.¹ In 2023, the Food and Drug Administration approved 2 products designed to prevent RSV LRTD in all infants: a bivalent, RSV prefusion F vaccination (RSVpreF; ABRYSV0, Pfizer) administered during pregnancy² and a long-acting monoclonal antibody (nirsevimab, BEYFORTUS, AstraZeneca) administered to infants at birth or at the beginning of their first RSV season.³ In clinical trials, each product demonstrated high efficacy in preventing LRTD caused by RSV in infants.^{4,5} Real-world effectiveness data have also shown that both maternal RSV vaccination⁶ and infant nirsevimab immunoprophylaxis^{7,8} are highly effective in preventing RSV LRTD associated with infant hospitalization and severe disease.

To prevent severe RSV disease in infants, the American Academy of Pediatrics (AAP) currently recommends either maternal RSV vaccination or infant immunization with nirsevimab or the recently approved clesrovimab (Enflonsia, Merck).^{9,10} Most infants do not need protection via both maternal vaccine and infant immunization, although there are certain circumstances when the benefit of administering both products should be considered. These circumstances include infants born less than 14 days after maternal RSV vaccination; infants born to mothers who may have a sub-optimal immune response to vaccination (eg, persons living with HIV infection); infants with accelerated loss of maternal antibodies, such as infants undergoing cardiopulmonary bypass; and infants with increased risk for severe RSV disease.² Infants born to mothers with unknown maternal vaccination history or those who received maternal vaccination during a previous pregnancy should also receive RSV monoclonal antibody. Thus, there is a need to understand the safety and tolerability of exposure to both maternal RSV vaccination and infant monoclonal antibody and to characterize the magnitude and durability of immune responses following exposure to both products. The purpose of this study was to evaluate the safety and immunology of maternal RSV vaccination and infant nirsevimab immunization administered alone or sequentially in mothers and infants. We hypothesized that both products would be safe and that differences in immunogenicity would be observed.

METHODS

Trial Design and Participants

A prospective, randomized, open-label, phase 4 study was conducted at 8 US sites. Mother-infant pairs were randomized 1:1:1:1 during pregnancy into 4 groups: group 1A, maternal RSVpreF vaccine alone; group 1B, maternal RSVpreF vaccine/infant nirsevimab at birth; group 1C, maternal RSVpreF vaccine/infant nirsevimab at 3 months; or group 2, infant nirsevimab alone at birth. Randomization

was performed in blocks by site using the enrollment module of AdvantageEDC software managed by the EMMES Corporation (Rockville, MD). Mothers with uncomplicated singleton pregnancies at 32 0/7 to 36 6/7 weeks gestational age, inclusive, were eligible to enroll. Full eligibility criteria are included in the Supplementary Material. Enrollments occurred from September 19, 2024, through March 28, 2025, and mother-infant pairs remain in follow-up for 12 months after delivery to ascertain safety, infant tolerability, and the magnitude and durability of RSV binding and neutralizing antibodies (nAbs) in mothers and infants. Study visit windows were defined per protocol (Supplementary Figure 1; Supplementary Table 1), and visits substantially out of window were excluded from analysis. The results presented in this report represent an interim analysis of data collected through May 15, 2025, which was the study's only preplanned interim analysis and included up to 4-month infant safety follow-up. All maternal participants provided written informed consent for themselves and their infant. This study was approved by Advarra institutional review board and posted on clinicaltrials.gov (NCT06551506) on August 13, 2024.

Data Collection

Following enrollment, maternal demographic, clinical, and obstetrical information; concomitant medications; and vaccines were collected through participant questionnaire, and baseline blood was collected. Participant race and ethnicity were collected, given known disparities in access to care and outcomes.^{11,12} Eligible participants were randomized, and those in group 1 received maternal RSVpreF vaccine (ABRYSV0), which was stored, handled, and administered according to the package insert. On delivery of the infant, cord blood (or infant blood if cord blood was not available), maternal blood, and optional breast milk samples were collected. Infant demographic information, medical history, pregnancy outcome, neonatal data, concomitant medications, and vaccinations administered within ± 7 days of study product were collected through questionnaire or medical record abstraction. A physical examination was performed, including measurement of infant weight, length, and head circumference. After confirming infant eligibility and maternal consent to continue in the study, infants randomized to groups 1B or 2 received nirsevimab (BEYFORTUS) within 7 days of delivery. Infants randomized to group 1C received nirsevimab on day 91 (± 7 days). Nirsevimab was stored according to the package insert, dosed per weight, and administered intramuscularly. Following nirsevimab administration, solicited local and systemic adverse reactions were collected through 7 days postimmunization. Mothers and infants were seen on days 43 and 91. At these visits, blood was collected from mothers and infants, and optional breast milk was collected to ascertain antibody responses. Breast milk was not analyzed for

this interim analysis but will be included in a future final study report. To limit the number of blood draws, mothers-infant pairs were randomized to have blood collection on either day 43 or day 91. Safety phone calls occurred on days 31 and 121. Maternal serious adverse events (SAEs) were collected through day 31. Infant medically attended adverse events (MAAEs) and unsolicited, grade 3 or higher adverse events (AEs) were collected for 30 days following each potential nirsevimab administration (ie, birth through day 31 and day 91 through day 121). Infant SAEs are being collected through day 181.

Validated RSV-A and -B Microneutralization Assays

Validated RSV-A and -B microneutralization assays were performed using collected sera. Heat-inactivated sera were analyzed for nAb activity against RSV/A/Tracy (GA1 genotype) and RSV/B/18537 (GB1 genotype) in HEp-2 cells as previously described^{13,14} (Supplementary Methods). All nAb titers were normalized based on an internal reference standard and the World Health Organization (WHO) International Standard. The internal reference standard had been benchmarked to the WHO 1st International Standard for Antiserum to RSV.^{15,16} The lower limit of detection of nAb titer in IU/mL was 12.0 and 7.0 against RSV/A/Tracy and RSV/B/18537, respectively. Although no threshold of protection against RSV disease has been established, infant nAb titers at delivery in the phase 2 study of RSVpreF vaccine were geometric mean titer (GMT) 28 051 IU/mL against RSV-A and 35 326 IU/mL against RSV-B.¹⁷ Similarly, infant nAb titers at delivery in the phase 3 trial of nirsevimab had GMT 19 737 IU/mL

against RSV-A at the earliest time point postadministration (day 31).¹⁸

Statistical Analyses

The safety analysis population included all mother-infant pairs who received either 1 or both study products. The primary immunogenicity population consisted of mother-infant pairs who received study product(s) and had at least 1 serum sample collected from the infant. Participants were analyzed as treated. This included participants who were treated as randomized and those who received different treatments from their assigned randomization (Supplementary Methods). Descriptive summaries of the safety data are presented for the safety analysis population, while descriptive summaries of immunogenicity data are presented for the primary immunogenicity population. GMTs of serum RSV-A and -B nAbs in IU/mL were calculated, along with 95% CIs, by study group, at each time point. The study was not powered for formal statistical comparisons between groups or time points (Supplementary Methods).

RESULTS

Study Participants

At the time of interim analysis, 181 mothers had enrolled at 8 US sites, and 180 mothers and 179 infants were included in the safety population (Supplementary Figure 2). The median age of mothers was 32.0 years (IQR, 7.0 years) and median gestational age was 33.3 weeks (IQR, 2.0 weeks) at the time of enrollment (Table 1). The median gestational age at the time of delivery was 39.0 weeks (IQR, 2.0 weeks),

TABLE 1. Baseline Maternal Characteristics					
Variable	Group 1A: Maternal RSV Vaccine Alone (n = 43)	Group 1B: Maternal RSV Vaccine Plus Infant Nirsevimab at Birth (n = 49)	Group 1C: Maternal RSV Vaccine Plus Infant Nirsevimab at 3 mo (n = 44)	Group 2: Infant Nirsevimab Alone at Birth (n = 44)	All Participants (n = 180)
Maternal age, median (IQR), y	32.0 (7.0)	33.0 (6.0)	32.0 (6.0)	34.0 (6.0)	32.0 (7.0)
Gestational age at enrollment, median (IQR), wk	33.3 (2.1)	33.6 (2.6)	33.3 (1.4)	33.2 (1.6)	33.3 (2.0)
Ethnicity, n (%)					
Not Hispanic or Latino	35 (81)	45 (92)	37 (84)	40 (91)	157 (87)
Hispanic or Latino	8 (19)	4 (8)	7 (16)	4 (9)	23 (13)
Race, n (%)					
American Indian or Alaska Native	0 (0)	0 (0)	1 (2)	0 (0)	1 (1)
Asian	3 (7)	8 (16)	8 (18)	7 (16)	26 (14)
Black or African American	5 (12)	8 (16)	6 (14)	5 (11)	24 (13)
White	29 (67)	30 (61)	29 (66)	26 (59)	114 (63)
Multiracial	3 (7)	2 (4)	0 (0)	6 (14)	11 (6)
Unknown	3 (7)	1 (2)	0 (0)	0 (0)	4 (2)
Abbreviation: RSV, respiratory syncytial virus.					

TABLE 2. Baseline Infant Characteristics					
Variable	Group 1A: Maternal RSV Vaccine Alone (n=43)	Group 1B: Maternal RSV Vaccine Plus Infant Nirsevimab at Birth (n=48)	Group 1C: Maternal RSV Vaccine Plus Infant Nirsevimab at 3 mos (n=44)	Group 2: Infant Nirsevimab Alone at Birth (n=44)	All Participants (n=179)
Gestational age at delivery, median (IQR), wk	39.0 (1.0)	39.0 (1.0)	39.0 (1.0)	39.0 (2.0)	39.0 (2.0)
Birth weight, median (IQR), kg	3.2200 (0.5950)	3.3625 (0.6105)	3.2975 (0.6375)	3.3150 (0.7400)	3.2900 (0.6500)
Sex, n (%)					
Male	21 (49)	27 (56)	22 (50)	21 (48)	91 (51)
Female	22 (51)	21 (44)	22 (50)	23 (52)	88 (49)
Ethnicity, n (%)					
Not Hispanic or Latino	34 (79)	44 (92)	34 (77)	38 (86)	150 (84)
Hispanic or Latino	8 (19)	4 (8)	7 (16)	5 (11)	24 (13)
Not reported/unknown	1 (2)	0 (0)	3 (7)	1 (2)	5 (3)
Race, n (%)					
Asian	3 (7)	7 (15)	7 (16)	5 (11)	22 (12)
Black or African American	6 (14)	7 (15)	5 (11)	8 (18)	26 (15)
White	25 (58)	29 (60)	29 (66)	22 (50)	105 (59)
Multiracial	7 (16)	4 (8)	2 (5)	9 (21)	22 (12)
Unknown	2 (5)	1 (2)	1 (2)	0 (0)	4 (2)
Days of delivery relative to maternal vaccination, n (%)					
<14 d	0 (0)	4 (8)	1 (2)	N/A	N/A
≥14 d	42 (98)	44 (92)	43 (98)	N/A	N/A
Gestational age at delivery, n (%), wk					
32–34	0 (0)	1 (2)	0 (0)	2 (5)	3 (2)
35–37	7 (16)	7 (15)	6 (14)	6 (14)	26 (15)
38–40	36 (84)	39 (81)	35 (80)	34 (77)	144 (80)
≥41	0 (0)	1 (2)	3 (7)	2 (5)	6 (3)
Abbreviations: N/A, not available; RSV, respiratory syncytial virus.					

and nearly all infants in group 1 were born at least 14 days after maternal vaccination (n = 129/135, 96%) (Table 2). The baseline characteristics of mothers and infants were similar between groups.

Safety and Reactogenicity

Overall, both study products were safe, whether administered alone or sequentially. There were no related maternal SAEs in either study group (Supplementary Table 2). For infants, solicited AEs following nirsevimab were mild to moderate and balanced across groups (groups 1B, 1C, and 2) (Figure 1, Supplementary Table 3). Local adverse reactions occurred in 17% of infants who received nirsevimab, the most common of which were injection site pain/tenderness (13%) and erythema (5%) (Figure 1). Systemic adverse reactions occurred in 60% of infants who received nirsevimab, the most common of which were sleepiness/fatigue (43%) and irritability/crying (43%) (Figure 1). Both local and systemic reactogenicity peaked on days 1 to 2 post-nirsevimab immunization. Local reactogenicity was resolved by day 6 in all infants (Supplementary Figure 3).

Mild systemic symptoms varied but appeared to be less common as the week progressed. Moderate solicited systemic symptoms declined in all groups over the 7-day collection period and resolved by day 7 in all infants (Supplementary Figure 4). Among infants, there were no related unsolicited grade 3 or higher AEs, and there were no related MAAEs or SAEs (Supplementary Table 3; Supplementary Figure 5).

Immunogenicity

In mothers, bivalent RSVpreF vaccination boosted maternal RSV-A and -B nAb titers 17.35-fold (95% CI, 14.48–20.79) and 23.98-fold (95% CI, 19.46–29.54), respectively, above baseline enrollment titers at delivery (Figure 2, A and B; Supplementary Figures 6 and 7; Supplementary Tables 4 and 5). Maternal RSV-A and -B nAb titers of 8547.0 (95% CI, 7077.8–10 321.3) and 12 185.6 (95% CI, 9650.9–15 386.0) IU/mL, respectively, were observed at delivery and remained durable through 3 months. Unvaccinated mothers in group 2 had minimal increases in antibody titers from delivery to 3 months (Supplementary Tables 4 and 5).

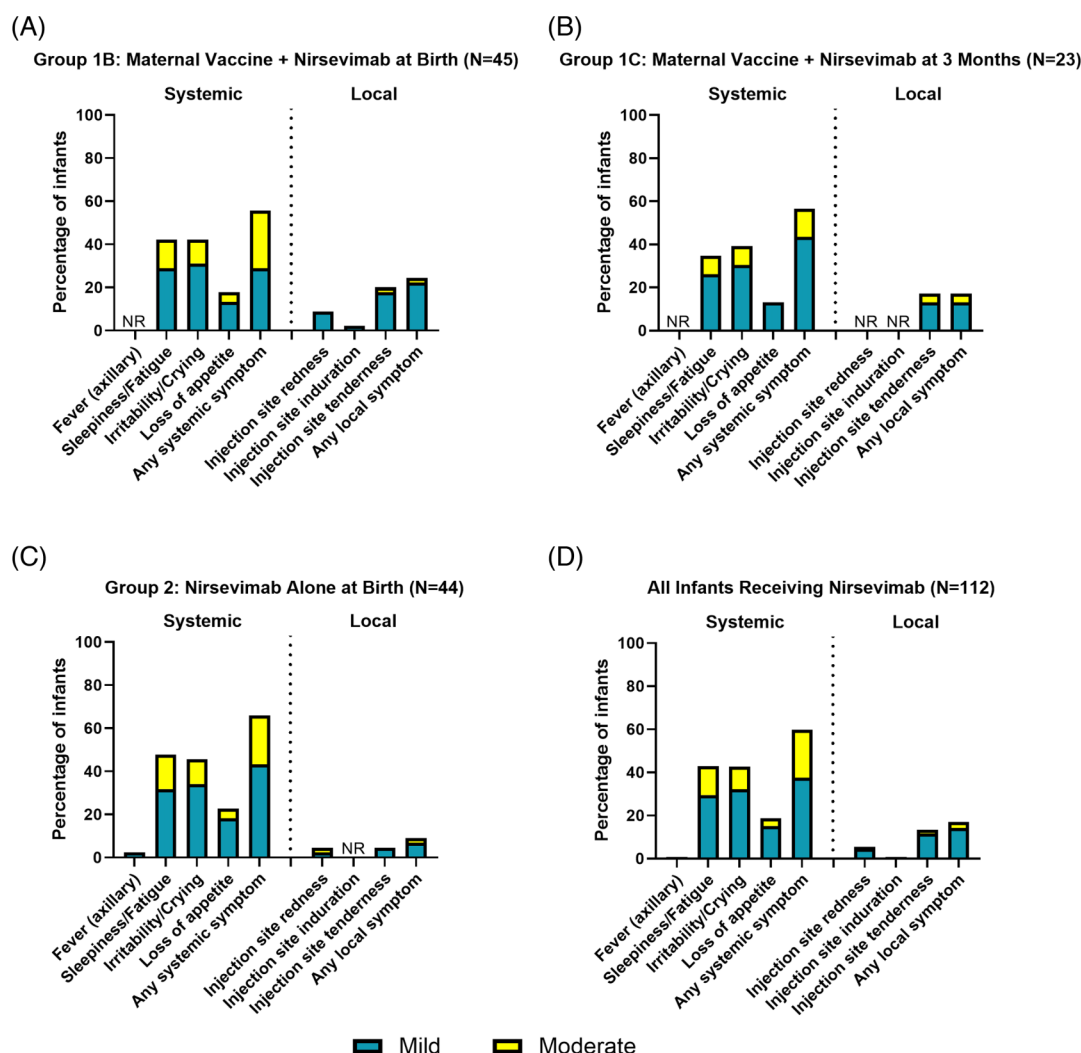


FIGURE 1.

Infant local and systemic reactogenicity following nirsevimab administration in groups 1B, 1C, and 2 through interim follow-up. Maximum severity of solicited events per infant participant by symptom and treatment group. Abbreviation: NR, no reaction.

In infants, maternal RSV vaccination resulted in highly elevated RSV-A and -B nAb titers at birth, with GMTs of 11 791.0 (95% CI, 9411.0–14 772.9) and 15 552.8 (95% CI, 11 175.5–21 644.5) IU/mL, respectively, in combined groups 1A and 1C (Supplementary Tables 6 and 7). Antibodies in these infants modestly waned by 3 months of life, although GMTs remained high at 1857.2 (95% CI, 1054.6–3270.5) for RSV-A and 2224.1 (95% CI, 1302.2–3798.6) IU/mL (Figure 2; Supplementary Figures 8 and 9; Supplementary Tables 6 and 7).

Compared with infants of mothers who received prenatal vaccination (group 1A and 1C), infants of mothers who did not receive maternal RSV vaccination (group 2) had lower RSV-A and -B nAb titers at birth (1109.0 [95% CI, 823.1–1494.3] and 1116.1 [95% CI, 783.6–1589.8] IU/mL), respectively. In group 2, administration of nirsevimab at

birth substantially increased RSV antibodies at day 43, with a geometric mean fold rise (GMFR) of 25.12 (95% CI, 13.34–47.32) to a peak titer of 28 980.6 IU/mL (95% CI, 20 360.1–41 251.0) for RSV-A and GMFR of 7.43 (95% CI, 3.73–14.78) to a peak titer of 7533.6 IU/mL (95% CI, 5174.3–10 968.6) for RSV-B. These antibodies were durable through 3 months of life.

Infants in group 1B, whose mothers received RSV vaccination and who were given nirsevimab at birth, had high RSV antibody titers at delivery that persisted through interim follow-up. At delivery, infants in group 1B had RSV-A nAb titers of 11 597.6 IU/mL (95% CI, 8550.1–15 731.5), which increased 3.53-fold (95% CI, 2.29–5.42) following nirsevimab administration to a peak of 41 052.5 IU/mL (95% CI, 33 797.2–47 703.0) at 6 weeks of life. These same infants had RSV-B nAb titers of 17 507.7 IU/mL

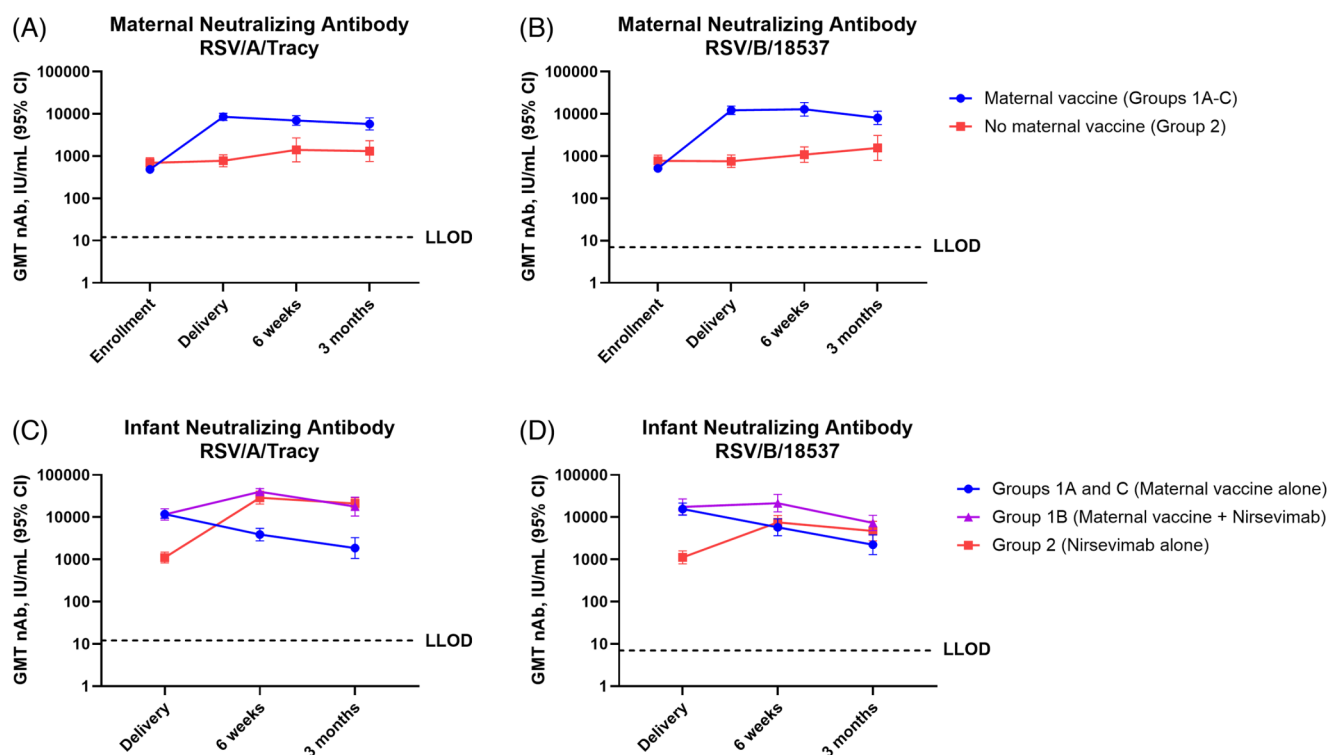


FIGURE 2.

RSV-A and -B neutralizing antibodies (nAbs) in mothers and infants, geometric mean titers (GMTs) with 95% CIs by time point and treatment group, primary immunogenicity population using RSV strains A/Tracy and B/18537.

Abbreviations: LLOD, lower limit of detection; RSV, respiratory syncytial virus.

(95% CI, 11 348.0–27 010.7) at delivery, which did not substantially increase following nirsevimab administration at 6 weeks of life, with a peak titer of 21 505.4 IU/mL (95% CI, 13 370.5–34 589.8).

The cord-to-maternal ratios (CMRs) of RSV-A and -B nAbs were similar, regardless of maternal RSV vaccination status or antigen specificity, reflecting efficient transplacental transfer of antibodies from mother to infant. Among those who received maternal RSV vaccination (group 1), RSV-A and -B CMRs were 1.36 (95% CI, 1.18–1.57) and 1.37 (95% CI, 1.15–1.62). Among those who did not receive maternal RSV vaccination (group 2), RSV-A and -B CMRs were 1.45 (95% CI, 1.25–1.70) and 1.48 (95% CI, 1.23–1.78) (Table 3).

DISCUSSION

We performed a prospective, phase 4, open-label, multicenter clinical trial at 8 US sites of mother-infant pairs randomized to receive maternal RSV vaccination, infant nirsevimab immunization, or both products to evaluate safety, infant tolerability, and the magnitude and durability of maternal and infant RSV antibodies. In this interim analysis, we found that both products alone and in combination were safe, with no related SAEs observed in mothers or infants.

Infants tolerated nirsevimab well, irrespective of study group. All local and systemic reactogenicity events were mild to moderate in severity, with no related grade 3 or higher unsolicited AEs or related MAAEs.

RSV vaccination substantially boosted RSV-A and -B nAb titers in mothers, which persisted through 3 months post-delivery. Transplacental transfer of RSV antibodies from mother to infant was high, with CMR higher than 1 irrespective of maternal vaccination status or antigen specificity. Importantly, infant RSV-A and -B nAbs were highly elevated following either maternal RSV vaccination, infant nirsevimab immunization, or sequential administration of both products. Following maternal vaccination alone, infants had peak RSV antibody titers at delivery, and these waned modestly by 3 months of life. In contrast, RSV antibody titers among infants in group 2 substantially increased following nirsevimab administration and persisted through 3 months of life. Although no threshold of protection against RSV disease has been well established, RSV neutralizing and prefusion F antibodies are known to correlate with protection against RSV disease in infants and to wane over the first months of life.^{19–23} While we observed subtle differences in RSV neutralization titers across groups and time points in our study, antibody titers following maternal vaccination

TABLE 3. CMRs with 95% CIs of RSV-A and -B Neutralizing Antibodies in Mother-Infant Pairs, Primary Immunogenicity Population		
Statistic	Group 1: Maternal RSV Vaccine (n = 114)	Group 2: No Maternal Vaccine (n = 38)
RSV/A/Tracy neutralizing antibodies		
Cord blood IU/mL (GMT)	11 272.5	1116.4
(95% CI)	(9379.1–13 548.2)	(815.4–1528.5)
Maternal serum IU/mL (GMT)	8283.9	767.9
(95% CI)	(6747.0–10 170.9)	(533.5–1105.3)
Transfer ratio (GMR)	1.36	1.45
(95% CI)	(1.18–1.57)	(1.25–1.70)
RSV/B/18537 neutralizing antibodies		
Cord blood IU/mL (GMT)	16 221.6	1108.3
(95% CI)	(12 307.9–21 379.9)	(756.8–1623.2)
Maternal serum IU/mL (GMT)	11 874.1	746.8
(95% CI)	(9265.1–15 217.6)	(522.3–1067.9)
Transfer ratio (GMR)	1.37	1.48
(95% CI)	(1.15–1.62)	(1.23–1.78)
Abbreviations: CMR, cord-to-maternal ratio; GMR, geometric mean transfer ratio; GMT, geometric mean titer; RSV, respiratory syncytial virus.		

and or infant nirsevimab were similar in magnitude to those observed in the clinical trials of RSVpreF and nirsevimab, which were highly efficacious.^{17,18}

The AAP recommends that infants aged younger than 8 months born during or entering their first respiratory virus season receive protection from RSV LRTD through either maternal vaccination or infant RSV monoclonal antibody (nirsevimab or clesrovimab),³ although each has different advantages. Maternal RSV vaccination, which is bivalent, elicits a response to both RSV-A and RSV-B; may provide benefit both mother and child; offers protection immediately on birth because of transplacental transfer of RSV antibodies; and may provide ongoing benefit through breast milk secretion of antibodies. Maternal vaccination avoids intramuscular injection in the infant and has been more accessible than RSV monoclonal antibodies to date. Disadvantages of maternal RSV vaccination include the limited timing (September through January) and the narrow gestational window of administration (32 0/7 through 36 6/7 weeks gestational age). Additionally, while maternal RSV vaccination is expected to confer protection to the infant through the first respiratory season, natural antibodies wane more quickly than monoclonal antibodies containing the stabilizing YTE substitutions.¹⁸ This difference in waning was also observed in our study and may impact durability of protection.

Advantages of infant nirsevimab administration include the reliable delivery of high-titer, long-half-life, RSV nAbs.^{3,18} The timing of administration can also be optimized to achieve peak effectiveness during the period of maximum exposure, including to certain high-risk infants during their second RSV season. Because the product is administered to the infant, it circumvents the possibility of any complication

during pregnancy. Disadvantages of nirsevimab include high cost, which can impede access; supply shortages, such as those encountered during the initial season²⁴; potential delays in administration if not dosed in the immediate newborn period; and the small risk of adverse reactions, such as hypersensitivity reactions or development of antidrug antibodies.^{5,25} It is also possible that RSV escape mutants may arise naturally or under selective pressure from an RSV mAb. In studies performed to date, the number of resistant strains detected has been small,²⁶ but viral surveillance is ongoing.

The indications for administering both maternal RSV vaccination and infant nirsevimab immunization are few. Prior studies have demonstrated that this approach is not cost-effective and is not likely needed to protect most infants. In our study, we found that either product alone or in combination provided high titers of nAbs to RSV in infants through 3 months of life. The infant nAb titers observed in our study were comparable with those observed in clinical trials of the 2 products, which have since been found to be highly effective.^{27,28} While modest differences in infant antibody titers, antigen specificity, and waning kinetics were observed in our study, we do not expect these to correspond with substantial differences in protection during these early months of life when infants are at highest risk of severe RSV. Nevertheless, we know that some infants will be exposed to both products, either inadvertently or for one of the clinical indications specified above. Reassuringly, our study found this approach to be safe and well tolerated.

This study has certain limitations. The relatively small sample size limits the ability to detect rare safety outcomes or make formal comparisons between groups. This study enrolled participants who had healthy, singleton pregnancies;

this sampling bias could limit applicability to real-world scenarios when both maternal vaccination and infant nirsevimab might be indicated. The study was not designed to ascertain differences in efficacy, and we did not perform surveillance for breakthrough clinical RSV infections. While seroconversion to RSV nucleoprotein will be included in planned analyses as a marker of RSV exposure, this was not included or adjusted for in this interim analysis. Finally, because there is not a universally accepted immune correlate of protection against RSV disease, interpretation of the serologic results limits our ability to extrapolate efficacy. Larger, population-based studies are needed to ascertain differences in real-world effectiveness. Nevertheless, strengths of this study include its prospective, randomized, multicenter study design that reduces bias and improves generalizability.

CONCLUSIONS

Maternal RSV vaccination and infant nirsevimab immunization, administered either alone or sequentially, were safe and provided high RSV-A and -B nAb titers in infants that persisted through 3 months after delivery. While most infants will not need to receive both products to be protected, our results suggest that maternal RSV vaccination

and infant nirsevimab immunization may be safely sequentially administered.

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ABBREVIATIONS

AE: adverse event
CMR: cord-to-maternal ratio
GMFR: geometric mean fold rise
GMT: geometric mean titer
LRTD: lower respiratory tract disease
MAAE: medically attended AE
nAb: neutralizing antibody
RSV: respiratory syncytial virus
RSVpreF: RSV prefusion F vaccination
SAE: serious AE
WHO: World Health Organization

Drs Eppes, Jones-Beatty, Badell, Quinn, Rick, and Forde made substantial contributions to acquisition of data. Drs Avadhanula, Telu, and Kunwar made substantial contributions to acquisition of data and analysis and interpretation of data. Drs Mu, Gao, Piper, Kim, Beresnev, and Cardemil made substantial contributions to conception and design and analysis and interpretation of data. Drs Flach, Posavad, and Miedema made substantial contributions to conception and design. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work. Deidentified individual participant data will not be made available. The ClinicalTrials.gov ID is NCT06551506: "The Immunology and Safety of Maternal RSV Vaccination (ABRYSVO), Infant Nirsevimab (BEYFORTUS) Immunization, or Both Products," first posted August 13, 2024.

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